cross_classified_models

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```r
library(arrow)
```

```
Warning: package 'arrow' was built under R version 4.4.3


Attaching package: 'arrow'

The following object is masked from 'package:utils':

    timestamp
```

```r
library(lme4)
```

```
Warning: package 'lme4' was built under R version 4.4.3

Loading required package: Matrix
```

```r
library(lmerTest)
```

```
Warning: package 'lmerTest' was built under R version 4.4.3


Attaching package: 'lmerTest'

The following object is masked from 'package:lme4':

    lmer
```

```
The following object is masked from 'package:stats':

    step
```

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(performance)
```

```
Warning: package 'performance' was built under R version 4.4.3
```

```r
library(broom.mixed)
```

```
Warning: package 'broom.mixed' was built under R version 4.4.3
```

```r
library(jtools) # new package
```

```
Warning: package 'jtools' was built under R version 4.4.3
```

```r
options(scipen = 999)
```

```r
# Setting repository path
#repo_path <- "/Users/namomac/Desktop/Moneyball_FC/inputs/"
repo_path <- "D:/repos/Moneyball_FC/inputs/"
# Reading data
dat <- read_parquet(paste0(repo_path, "player_df.parquet"))


# View the data
glimpse(dat)
```

```
Rows: 7,168
Columns: 27
$ season_year             <dbl> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013~
$ club_id                 <dbl> 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11~
$ player_id               <dbl> 132, 5293, 6710, 7451, 15185, 15378, 15799, 15~
$ yellow_cards            <dbl> 3, 1, 4, 8, 3, 1, 4, 5, 1, 2, 3, 0, 2, 3, 0, 2~
$ red_cards               <dbl> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ goals                   <dbl> 3, 0, 3, 6, 16, 0, 12, 0, 0, 0, 0, 0, 7, 16, 0~
$ assists                 <dbl> 1, 0, 0, 6, 12, 1, 13, 0, 0, 1, 0, 0, 5, 13, 0~
$ minutes_played          <dbl> 904, 90, 3838, 3795, 2610, 132, 4131, 3047, 90~
$ name                    <chr> "Tomas Rosicky", "Sébastien Squillaci", "Per M~
$ country_of_birth        <chr> "CSSR", "France", "Germany", "Spain", "Poland"~
$ country_of_citizenship  <chr> "Czech Republic", "France", "Germany", "Spain"~
$ date_of_birth           <dttm> 1980-10-04, 1980-08-11, 1984-09-29, 1982-03-2~
$ sub_position            <chr> "Attacking Midfield", "Centre-Back", "Centre-B~
$ position                <chr> "Midfield", "Defender", "Defender", "Midfield"~
$ height_in_cm            <dbl> 179, 185, 200, 177, 182, 172, 168, 183, 188, 1~
$ market_value_ave        <dbl> 2000000.0, 833333.3, 16000000.0, 10000000.0, 2~
$ Age                     <dbl> 33, 33, 29, 31, 28, 32, 29, 28, 29, 27, 30, 28~
$ Club                    <chr> "Arsenal", "Arsenal", "Arsenal", "Arsenal", "A~
$ Country                 <chr> "England", "England", "England", "England", "E~
$ year_ex_rate            <dbl> 1.33, 1.33, 1.33, 1.33, 1.33, 1.33, 1.33, 1.33~
$ market_value_ave_usd    <dbl> 2660000, 1108333, 21280000, 13300000, 30590000~
$ usd_revenue_2024        <dbl> 3581828.4, 1492428.5, 28654627.2, 17909142.0, ~
$ foreign_players         <dbl> 0.8214286, 0.8214286, 0.8214286, 0.8214286, 0.~
$ uefa_coefficient        <dbl> 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15~
$ country_of_birth_fixed  <chr> "Czechoslovak Socialist Republic", "France", "~
$ Region                  <chr> "Europe & Central Asia", "Europe & Central Asi~
$ citizen_of_club_country <chr> "No", "No", "No", "No", "No", "No", "No", "No"~
```

Data Structure

Level 1 (time): season_year: season-year within player Level 2 (player): name: players Level 3 (club): club_id: clubs

L1 time-varying: season_year, yellow_cards, red_cards, goals, assists, minutes_played L2 player-invariant: country_of_birth, sub_position, position, height_in_cm L3 club-invariant: season_year

seanson_year: 2013, 2014, 2015, 2016, 2017, 2018, 2020, 2021, 2022, 2023, 2024.

position: Attack, Defender, Goalkeeper, Midfield.

Region: East Asia & Pacific, Europe & Central Asia, Latin America & Caribbean, Middle East & North Africa, North America, Sub-Saharan Africa.

citizen_of_club_country: YES, NO.

```r
player_df <- dat %>%
  mutate(
    player_id = factor(player_id),
    club_id   = factor(club_id),
    position = factor(position),
    Region = factor(Region),
    citizen_of_club_country = factor(citizen_of_club_country),
    usd_revenue_2024_log = log(usd_revenue_2024),
    # convert season_year to 0, 1, 2, etc.
    season_year_c = season_year - min(season_year)
  )
```

- Check missing values.

```r
# Which variable has missing value
colSums(is.na(player_df))
```

```
            season_year                 club_id               player_id 
                      0                       0                       0 
           yellow_cards               red_cards                   goals 
                      0                       0                       0 
                assists          minutes_played                    name 
                      0                       0                       0 
       country_of_birth  country_of_citizenship           date_of_birth 
                    108                     103                       2 
           sub_position                position            height_in_cm 
                      0                       0                      36 
       market_value_ave                     Age                    Club 
                      0                       2                       0 
                Country            year_ex_rate    market_value_ave_usd 
                      0                       0                       0 
       usd_revenue_2024         foreign_players        uefa_coefficient 
                      0                       0                       0 
 country_of_birth_fixed                  Region citizen_of_club_country 
                    108                     108                     103 
   usd_revenue_2024_log           season_year_c 
                      0                       0 
```

```r
# Which observation has missing values
player_df %>%
  filter(!complete.cases(.))
```

```
# A tibble: 145 x 29
   season_year club_id player_id yellow_cards red_cards goals assists
         <dbl> <fct>   <fct>            <dbl>     <dbl> <dbl>   <dbl>
 1        2013 12      4311                 1         0     0       0
 2        2013 15      1073                 3         0     2       0
 3        2013 15      43274               12         0     3       1
 4        2013 15      51168                3         0     5       0
 5        2013 15      53173                7         0     3       0
 6        2013 16      58124                5         0     5       2
 7        2013 27      2514                10         0     9      11
 8        2013 46      28021                4         0     0       0
 9        2013 148     3476                 1         0     0       0
10        2013 148     4192                 0         0     0       0
# i 135 more rows
# i 22 more variables: minutes_played <dbl>, name <chr>,
#   country_of_birth <chr>, country_of_citizenship <chr>, date_of_birth <dttm>,
#   sub_position <chr>, position <fct>, height_in_cm <dbl>,
#   market_value_ave <dbl>, Age <dbl>, Club <chr>, Country <chr>,
#   year_ex_rate <dbl>, market_value_ave_usd <dbl>, usd_revenue_2024 <dbl>,
#   foreign_players <dbl>, uefa_coefficient <dbl>, ...
```

```r
# Exclude all missing values, only 7023 observations now
clean_player_df <- na.omit(player_df)
```

- Intercept only model: We want to model usd_revenue_2024 because market value is accounting for inflation and converted to 2024 US dollars.

Fit the model

Cross-classified

- The player went to different clubs across years, thus they are cross-clubs, not nested in one club.
- So we use (1 | club_id) + (1|player_id) here, not (1 | club_id) + (1|club_id : player_id) here, since the latter one is used only when player are nested in the clubs.

```r
clean_player_df %>%
  distinct(player_id, club_id) %>%
  count(player_id, name = "num_clubs") %>%
  filter(num_clubs > 1) %>%
  arrange(desc(num_clubs))
```

```
# A tibble: 441 x 2
   player_id num_clubs
   <fct>         <int>
 1 182712            7
 2 45320             5
 3 55735             5
 4 119296            5
 5 282429            5
 6 27389             4
 7 35564             4
 8 37389             4
 9 37666             4
10 39153             4
# i 431 more rows
```

We will now add a fixed effect for year and allow it to vary by club.

- Recall that year has been centered at 0, so that 2013=0, 2024 = 11. This is because `lmer` struggles when predictors are on vastly different scales than the outcome. If `season_year` is recorded as "2020, 2021, 2022," these numbers are huge compared to the log-revenue variance. This causes the gradient calculation to fail.

```r
fixed_year <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
      (1|player_id) + (1|club_id), ,
  data = clean_player_df, REML = TRUE)

summ(fixed_year)
```

- We can see that player_id accounts for about 76% of market value while club_id accounts for 3%.

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20052.80
BIC	20087.09
Pseudo-R^2 (fixed effects)	0.00
Pseudo-R^2 (total)	0.79

Fixed Effects

	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.99	0.07	221.31	32.91	0.00
season_year_c	-0.03	0.00	-8.11	6524.03	0.00

p values calculated using Satterthwaite d.f.

Random Effects

Group	Parameter	Std. Dev.
player_id	(Intercept)	1.34
club_id	(Intercept)	0.28
Residual		0.71

Grouping Variables

Group	# groups	ICC
player_id	2159	0.76
club_id	21	0.03

- we also fit a random slope for year to compare.
    - To estimate a random slope `(1 + year_centered | player_id)`, the model needs multiple data points for *every* player to calculate their individual trajectory. But many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the "player slope" from the "residual error," causing these convergence issues. Clubs, however, have many playuers and many years of historical market value data so a random slope for clubs is more robust to fit.

```
random_year <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
```

```r
    season_year_c +
    # random effects
      (1|player_id) + (1+season_year_c|club_id),
  data = clean_player_df, REML = TRUE)

summ(random_year)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19993.96
BIC	20041.96
Pseudo-R^2 (fixed effects)	0.00
Pseudo-R^2 (total)	0.80

Fixed Effects

	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.98	0.09	187.65	24.67	0.00
season_year_c	-0.03	0.01	-3.17	20.11	0.00

p values calculated using Satterthwaite d.f.

Random Effects

Group	Parameter	Std. Dev.
player_id	(Intercept)	1.35
club_id	(Intercept)	0.35
club_id	season_year_c	0.04
Residual		0.70

Grouping Variables

Group	# groups	ICC
player_id	2159	0.75
club_id	21	0.05

- AIC and BIC values are lower with the random slope of year.

```r
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced <- as.numeric(logLik(reduced_model))
  ll_full    <- as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  # Formula: -2 * (LL_reduced - LL_full)
  lrt_stat <- -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  # This tells us how many params we added
  df_reduced <- attr(logLik(reduced_model), "df")
  df_full    <- attr(logLik(full_model), "df")
  df_diff    <- df_full - df_reduced

  # 4. Calculate P-Value based on the difference
  # If we added a slope + covariance, df_diff should be 2.
  # The mixture is 0.5 * ChiSq(1) + 0.5 * ChiSq(2)

  if (df_diff == 2) {
    p_val <- 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
             0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

    msg <- "Mixture of ChiSq(1) and ChiSq(2)"

  } else if (df_diff == 1) {
    # If we added a parameter with NO covariance (e.g. diag structure)
    p_val <- 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
             0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

    msg <- "Mixture of ChiSq(0) and ChiSq(1)"

  } else {
    # Fallback for non-standard comparisons
    p_val <- pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
    msg <- paste0("Standard ChiSq(", df_diff, ")")
  }

  # Output
  cat("LRT Statistic:", round(lrt_stat, 4), "\n")
  cat("Parameter Diff:", df_diff, "\n")
```

```
  cat("Distribution:  ", msg, "\n")
  cat("P-Value:       ", format.pval(p_val, eps = .0001), "\n")
}

# Run the comparison
calculate_lrt_mixture(fixed_year, random_year)
```

```
LRT Statistic: 62.841
Parameter Diff: 2
Distribution:   Mixture of ChiSq(1) and ChiSq(2)
P-Value:        < 0.0001
```

- We use a helper function to print the p-value for the LRT difference, and find a p-value <.0001 and we reject the Null, (with .5 × 2(1) + .5 × 2(2 mixture) and conclude that there is strong evidence for club variation in market value trajectories and therefore we should keep this random slope in the model.

```
compare_performance(fixed_year, random_year, metrics = c("AIC", "BIC"))
```

```
Some of the nested models seem to be identical and probably only vary in
  their random effects.

# Comparison of Model Performance Indices

Name        |           Model |   AIC (weights) |   BIC (weights)
-----------------------------------------------------------------
fixed_year  | lmerModLmerTest | 20040.0 (<.001) | 20074.3 (<.001)
random_year | lmerModLmerTest | 19982.9 (>.999) | 20030.9 (>.999)
```

Adding first level player and club predictors

- We first want to group center minutes played for each for each club by year.

```
clean_player_df <- clean_player_df |>
  group_by(season_year_c, club_id) |>
  mutate(minutes_played_c = minutes_played - mean(minutes_played, na.rm=TRUE)) |>
  ungroup()
```

- We may want to use age at start of career and center it. We also need to account for the quadratic relationship between age and market value.
    - We include the first age observed for each player, centered, and squared and find a slightly better fit based on AIC and BIC.

```
clean_player_df <- clean_player_df |>
  group_by(player_id) |>
  mutate(first_age = first(Age)) |>
  ungroup() |>
  mutate(first_age_c = (first_age - mean(first_age, na.rm=TRUE))^2 )
```

```
player_varying_predictors <-  lmer(
  usd_revenue_2024_log ~ season_year_c +
    # first level predictors
    yellow_cards + red_cards + goals +
    assists + minutes_played + first_age_c +
    # random effects
      (1|player_id) + (season_year_c|club_id),
  data = clean_player_df, REML = TRUE)
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
summ(player_varying_predictors)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17537.57
BIC	17626.71
Pseudo-R^2 (fixed effects)	0.27
Pseudo-R^2 (total)	0.79

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.62	0.09	177.37	24.73	0.00
season_year_c	-0.01	0.01	-1.41	20.04	0.17
yellow_cards	0.01	0.00	1.78	5665.19	0.07
red_cards	-0.02	0.03	-0.64	4657.69	0.52
goals	0.02	0.00	6.56	6011.90	0.00
assists	0.01	0.00	3.34	5287.38	0.00
minutes_played	0.00	0.00	28.79	6048.10	0.00
first_age_c	-0.01	0.00	-17.13	1788.25	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.93
club_id	(Intercept)	0.37
club_id	season_year_c	0.03
Residual		0.62

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.62
club_id	21	0.10

We next add first level club predictors

- We want to center player height in inches across years. We also want to center uefa coefficients by year.

```
clean_player_df <- clean_player_df |>
  group_by(season_year) |>
  mutate(height_in_cm_c = height_in_cm - mean(height_in_cm, na.rm=TRUE),
         uefa_coefficient_c = uefa_coefficient -
           mean(uefa_coefficient)) |>
  ungroup()
```

```r
player_club_varying_predictors <-  lmer(
  usd_revenue_2024_log ~ season_year_c +
    # first level player predictors
    yellow_cards + red_cards + goals +
    assists + minutes_played + first_age_c +
    # first level club predictors
    height_in_cm_c + foreign_players + uefa_coefficient_c +
    # second level predictors
    position  + Region + citizen_of_club_country+
      # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE)
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```r
summ(player_club_varying_predictors)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17405.08
BIC	17562.79
Pseudo-R^2 (fixed effects)	0.35
Pseudo-R^2 (total)	0.79

Adding level 2 predictors

- We want to center height in inches across years.

```r
clean_player_df <- clean_player_df |>
  group_by(season_year_c) |>
  mutate(height_in_cm_c = height_in_cm - mean(height_in_cm, na.rm=TRUE)) |>
  ungroup()
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.37	0.22	69.17	1189.32	0.00
season_year_c	-0.01	0.00	-3.26	6946.34	0.00
yellow_cards	0.00	0.00	0.91	5571.98	0.36
red_cards	-0.02	0.03	-0.83	4692.24	0.41
goals	0.01	0.00	5.13	5728.90	0.00
assists	0.01	0.00	3.18	5171.68	0.00
minutes_played	0.00	0.00	30.09	5863.72	0.00
first_age_c	-0.01	0.00	-15.23	1764.53	0.00
height_in_cm_c	0.01	0.00	3.23	1579.18	0.00
foreign_players	0.39	0.12	3.41	5432.77	0.00
uefa_coefficient_c	0.00	0.00	1.16	5094.85	0.25
positionDefender	-0.30	0.06	-5.29	1752.45	0.00
positionGoalkeeper	-0.81	0.09	-8.71	1670.75	0.00
positionMidfield	-0.14	0.06	-2.41	1670.66	0.02
RegionEurope & Central Asia	0.27	0.20	1.38	1643.89	0.17
RegionLatin America & Caribbean	0.49	0.20	2.44	1632.01	0.01
RegionMiddle East & North Africa	0.31	0.30	1.03	1608.24	0.30
RegionNorth America	0.21	0.29	0.73	1664.56	0.46
RegionSub-Saharan Africa	0.31	0.22	1.41	1631.79	0.16
citizen_of_club_countryYes	-0.37	0.04	-10.13	5144.97	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.86
club_id	(Intercept)	0.30
Residual		0.63

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.60
club_id	21	0.07

```r
player_club_predictors <-  lmer(
  usd_revenue_2024_log ~ season_year_c +
    # first level player predictors
    yellow_cards + red_cards + goals +
    assists + minutes_played + first_age_c +
    # first level club predictors
    height_in_cm_c + foreign_players + uefa_coefficient_c +
    # second level predictors
    position  + Region + citizen_of_club_country+
      # random effects
      (1|player_id) + (season_year_c|club_id),
  data = clean_player_df, REML = TRUE)
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
Model failed to converge with max|grad| = 0.0100803 (tol = 0.002, component 1)
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```r
summ(player_club_predictors)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17353.33
BIC	17524.75
Pseudo-R^2 (fixed effects)	0.34
Pseudo-R^2 (total)	0.79

```r
export_summs(random_year, player_varying_predictors,
             player_club_varying_predictors, player_club_predictors,
             exp=TRUE, digits=3, r.squared=FALSE, confint = TRUE,
             error_format = "[{conf.low}, {conf.high}]",
             model.names = c("M1", "M2", "M3", "M4"))
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.34	0.23	66.39	735.26	0.00
season_year_c	-0.01	0.01	-1.31	20.01	0.21
yellow_cards	0.00	0.00	0.95	5529.50	0.34
red_cards	-0.02	0.03	-0.67	4658.85	0.50
goals	0.01	0.00	4.88	5695.59	0.00
assists	0.01	0.00	3.06	5133.36	0.00
minutes_played	0.00	0.00	30.19	5807.22	0.00
first_age_c	-0.01	0.00	-15.16	1765.83	0.00
height_in_cm_c	0.01	0.00	3.37	1577.83	0.00
foreign_players	0.43	0.13	3.39	3065.77	0.00
uefa_coefficient_c	-0.00	0.00	-0.29	4787.48	0.77
positionDefender	-0.31	0.06	-5.43	1743.05	0.00
positionGoalkeeper	-0.83	0.09	-8.92	1664.61	0.00
positionMidfield	-0.14	0.06	-2.49	1663.04	0.01
RegionEurope & Central Asia	0.29	0.20	1.45	1643.07	0.15
RegionLatin America & Caribbean	0.50	0.20	2.49	1631.68	0.01
RegionMiddle East & North Africa	0.33	0.30	1.10	1606.38	0.27
RegionNorth America	0.24	0.29	0.83	1660.41	0.41
RegionSub-Saharan Africa	0.31	0.22	1.43	1631.28	0.15
citizen_of_club_countryYes	-0.37	0.04	-10.14	5147.82	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.87
club_id	(Intercept)	0.38
club_id	season_year_c	0.03
Residual		0.63

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.59
club_id	21	0.11

	M1	M2	M3	
	8717408.041 ***	6075985.905 ***	4733593.657 ***	
	[7377275.862, 10300984.316]	[5112789.151, 7220638.994]	[3062271.210, 7317088.321]	[292118
	0.971 **	0.989	0.989 **	
	[0.954, 0.989]	[0.974, 1.004]	[0.983, 0.996]	
		1.007	1.004	
		[0.999, 1.015]	[0.996, 1.012]	
		0.981	0.975	
		[0.925, 1.040]	[0.919, 1.035]	
		1.017 ***	1.014 ***	
		[1.012, 1.022]	[1.008, 1.019]	
		1.012 ***	1.012 **	
		[1.005, 1.020]	[1.005, 1.019]	
		1.000 ***	1.000 ***	
		[1.000, 1.000]	[1.000, 1.000]	
		0.986 ***	0.988 ***	
		[0.984, 0.987]	[0.986, 0.989]	
			1.011 **	
			[1.004, 1.018]	
			1.483 ***	
			[1.182, 1.859]	
			1.002	
			[0.999, 1.005]	
			0.741 ***	
			[0.663, 0.828]	
			0.447 ***	
			[0.373, 0.536]	
			0.873 *	
			[0.782, 0.975]	
ral Asia			1.313	
			[0.892, 1.932]	
& Caribbean			1.634 *	